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Oefeningenreeks 2E nr. 4.7

$$\begin{aligned} & \lim_{x \rightarrow \infty} \left(\frac{ax+b}{ax+c} \right)^x \\ &= \lim_{x \rightarrow \infty} \left(\frac{ax+c+d}{ax+c} \right)^x \quad (\text{met } c+d=b) \\ &= \lim_{x \rightarrow \infty} \left(1 + \frac{d}{ax+c} \right)^x \\ &= \lim_{x \rightarrow \infty} \left(\left(1 + \frac{d}{ax+c} \right)^{\frac{ax+c}{d} \cdot \frac{d}{ax+c}} \right)^x \\ &= e^{\lim_{x \rightarrow \infty} \frac{dx}{ax+c}} \\ &= e^{\frac{d}{a}} \\ &= e^{\frac{b-c}{a}} \end{aligned}$$

References